

JavaScript Loops

What is a Loop?

A **loop** is a programming structure that allows you to execute the same block of code multiple times until a specified condition becomes false.

Instead of writing the same code repeatedly, you can use loops to automate repetitive tasks.

Why Do We Use Loops?

Loops help us:

- Execute code multiple times
- Iterate through arrays
- Process data efficiently
- Reduce duplicate code
- Improve readability

Types of Loops

1. for Loop
2. while Loop
3. do...while Loop
4. for...of Loop
5. for...in Loop

1. for Loop

Definition

The `for` loop is used when the number of iterations is known.

Syntax

```
for (initialization; condition; update) {  
    // code  
}
```

Parts of a for Loop

| Part | Purpose | ||-| | Initialization | Runs once before the loop starts | | Condition | Checked before every iteration | | Update | Runs after each iteration |

Example

```
for(let i = 1; i <= 5; i++){  
    console.log(i);  
}
```

Output

```
1  
2  
3  
4  
5
```

Step by Step

Iteration 1

```
i = 1  
  
1 <= 5  
  
true  
  
Print 1  
  
i++
```

Iteration 2

```
i = 2  
  
Print 2
```

...

Iteration 5

```
Print 5
```

Iteration 6

```
i = 6  
  
6 <= 5  
  
false  
  
Loop Stops
```

Real Project Example

Display product numbers.

```
for(let i = 1; i <= 10; i++){  
  console.log(`Product ${i}`);  
}
```

2. while Loop

Definition

The `while` loop executes as long as the condition remains true.

Syntax

```
while(condition){  
  // code  
}
```

Example

```
let i = 1;  
  
while(i <= 5){  
  console.log(i);  
  i++;  
}
```

Output

```
1  
2
```

```
3  
4  
5
```

Real Project Example

Loading data from an API until all pages are fetched.

3. do...while Loop

Definition

The `do...while` loop executes the code **at least once**, even if the condition is false.

Syntax

```
do{  
  
}while(condition);
```

Example

```
let i = 10;  
  
do{  
    console.log(i);  
    i++;  
}while(i < 5);
```

Output

10

Why?

The code inside `do` executes first.

Then JavaScript checks the condition.

4. for...of Loop

Definition

The `for...of` loop is used to iterate over iterable objects like arrays and strings.

Example

```
const fruits = ["Apple", "Mango", "Orange"];

for(const fruit of fruits){
  console.log(fruit);
}
```

Output

```
Apple
Mango
Orange
```

String Example

```
const name = "Rokon";

for(const letter of name){
  console.log(letter);
}
```

Output

```
R
o
k
o
n
```

Real Project Example

Display all products.

```
for(const product of products){
  console.log(product.name);
}
```

5. for...in Loop

Definition

The `for...in` loop is used to iterate over object properties.

Example

```
const student = {  
  name: "Rokon",  
  age: 23,  
  country: "Bangladesh"  
};  
  
for(const key in student){  
  console.log(key);  
}
```

Output

```
name  
age  
country
```

Get Values

```
for(const key in student){  
  console.log(student[key]);  
}
```

Output

```
Rokon  
23  
Bangladesh
```

Loop Control Statements

break

Stops the loop immediately.

Example

```
for(let i=1;i<=10;i++){  
  
    if(i===5){  
        break;  
    }  
  
    console.log(i);  
  
}
```

Output

```
1  
2  
3  
4
```

continue

Skips the current iteration.

Example

```
for(let i=1;i<=5;i++){  
  
    if(i===3){  
        continue;  
    }  
  
}
```

```
    console.log(i);  
  
}
```

Output

```
1  
2  
4  
5
```

Nested Loop

A loop inside another loop.

```
for(let row=1; row<=3; row++){  
    for(let col=1; col<=3; col++){  
        console.log(row,col);  
    }  
}
```

Common Loop Examples

Print Numbers

```
for(let i=1;i<=100;i++){  
    console.log(i);  
}
```

Print Even Numbers

```
for(let i=2;i<=20;i+=2){  
  console.log(i);  
}
```

Print Odd Numbers

```
for(let i=1;i<=20;i+=2){  
  console.log(i);  
}
```

Sum of Numbers

```
let sum = 0;  
  
for(let i=1;i<=5;i++){  
  sum += i;  
}  
  
console.log(sum);
```

Output

```
15
```

Multiplication Table

```
let number = 5;

for(let i=1;i<=10;i++){
  console.log(`${number} x ${i} = ${number*i}`);
}
```

Which Loop Should You Use?

| Loop | Best Use | ||-----|-----| | for | Known number of iterations | | while | Unknown number of iterations | | do...while | Run at least once | | for...of | Arrays & Strings | | for...in | Objects |

Best Practices

Use `for` when you know how many times to loop.

Use `for...of` for arrays.

Use `for...in` for objects.

Avoid infinite loops.

Common Mistakes

Forgetting Update



```
let i = 1;

while(i<=5){

  console.log(i);

}
```

Infinite Loop

Wrong Condition

```
for(let i=1;i>=10;i++){  
}
```

The loop never executes.

Interview Questions

What is a loop?

A loop is a control structure that repeatedly executes a block of code while a condition remains true.

Difference between for and while

| for | while | ||----|-----| | Known iterations | Unknown iterations | | Initialization inside loop | Initialization outside loop | | Compact syntax | More flexible |

Difference between for...of and for...in

for...of	for...in
Arrays	Objects
Returns values	Returns keys

Summary

Loop	Used For
for	Fixed iterations
while	Unknown iterations

Loop	Used For
do...while	Executes at least once
for...of	Arrays & Strings
for...in	Objects
break	Exit loop
continue	Skip current iteration